

Curvature Polarization Transport Gravity: Aperture-Resolved Active Gates in Galaxy Clusters

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Abstract

This paper develops a galaxy-cluster extension of Curvature Polarization Transport Gravity (CPTG) in which the response of a selected overdensity aperture is determined by baryonic loading, thermal support, curvature-branch expansion, and an active structural gate. At a declared aperture R_Δ , the calculation maps redshift, overdensity, aperture radius, baryonic mass, and support temperature into a loading variable F_Δ , a virial support scale $T_{\text{vir},\Delta}$, a gate variable Q_C , a response ratio y_Δ , a structural mode $\mathcal{N}_{C,\Delta}$, and a predicted cluster mass $M_{\text{CPTG},\Delta}$.

The central result is an aperture-resolved active-gate law. High- Q_C apertures approach closure with $y_\Delta \simeq 1$, while low- Q_C apertures enter suppressed response states with $y_\Delta < 1$. The structural mode $\mathcal{N}_{C,\Delta} = y_\Delta N_{C,0}/Q_C^2$ measures the response load carried by the active curvature branch at the selected aperture. Applied to ACCEPT radial-profile diagnostics, the law orders outer-profile rows into closure-stable, watch, suppressed, and strongly suppressed response classes. Applied to X-COP same-aperture R_{500} products, the law returns a median CPTG-to-hydrostatic mass ratio of approximately 0.989 across the tested 12-cluster sample, with a median absolute fractional difference of approximately 1.08%. The X-COP overdensity ladder shows a systematic transition from inner closure at R_{2500} to outer active-gate sensitivity at R_{500} . These results define a practical cluster-scale CPTG active-gate framework for comparing baryonic loading, thermal support, aperture structure, and predicted cluster mass within a single geometric response law.

1 Introduction

Galaxy clusters provide a natural scale for testing how a geometric gravity theory behaves when baryonic matter is distributed through an extended hot plasma rather than concentrated in a stellar disk. A cluster is not characterized by a single radius, a single acceleration, or a single local support channel. It is an aperture-resolved object whose gas mass, stellar mass, thermal support, pressure structure, and inferred gravitating mass depend on the overdensity radius at which the system is evaluated. A useful cluster-scale CPTG [4, 5] law must therefore be explicit about aperture.

This paper develops a cluster active-gate law for CPTG. The law is not written as a direct rescaling of a galaxy rotation formula. It is written as an aperture calculation. At each selected overdensity radius R_Δ , the observed or supplied cluster quantities define a baryonic seed and a support state. Those quantities are then passed through an active curvature-response gate. The gate determines whether the aperture is closure-stable, in a transition band, suppressed, or strongly suppressed.

The active-gate approach is designed around a simple practical chain:

$$\{z, \Delta, R_\Delta, M_b(< R_\Delta), T_{\text{gate}}\} \longrightarrow \{F_\Delta, Q_C, y_\Delta, \mathcal{N}_{C,\Delta}, M_{\text{CPTG},\Delta}\}. \quad (1)$$

The input side contains the redshift, overdensity aperture, radius, baryonic content, and support temperature. The output side contains the baryonic loading, active-gate state, response ratio, structural mode, and predicted cluster mass. This makes the law usable as a practical cluster tool: choose an aperture, supply the baryonic and thermal support information for that aperture, solve the active-gate equation, and compare the predicted CPTG mass to an observed mass proxy at the same aperture.

Two observational data streams are used as diagnostic applications. ACCEPT provides Chandra-based intracluster-medium entropy and temperature profile information for a large archival cluster sample [1, 2]. X-COP provides high-quality X-ray and Sunyaev-Zeldovich cluster products designed for cluster outskirts and hydrostatic mass reconstruction [3]. The ACCEPT use in this paper is a radial-profile gate-ordering diagnostic. The X-COP use is a same-aperture mass and aperture-ladder test.

The structure of the paper is as follows. Section 2 defines the parameters, input quantities, unit conventions, and aperture rules. Section 3 gives the active-gate equations and explains the role of each term. Section 4 introduces the cluster structural mode $\mathcal{N}_{C,\Delta}$. Section 5 describes the numerical solution and gate classifications. Section 6 gives a standalone calculation recipe with worked examples. Sections 7 and 8 present the ACCEPT and X-COP results separately. Section 9 summarizes the cluster-scale interpretation, and Section 10 gives the conclusion.

2 Parameters and Inputs

The cluster law is evaluated at a declared overdensity aperture. The aperture label Δ identifies the mean enclosed density convention relative to the critical density branch used by the calculation. The corresponding radius is R_Δ . All baryonic masses and comparison masses must be evaluated inside the same aperture.

The primary input set is

$$\mathcal{I}_\Delta = \{z, \Delta, R_\Delta, M_{\text{gas}}(< R_\Delta), M_\star(< R_\Delta), M_{\text{ICL}}(< R_\Delta), T_{\text{gate}}\}. \quad (2)$$

Here z is the cluster redshift, R_Δ is the aperture radius, M_{gas} is the enclosed gas mass, $M_\star$ is the enclosed stellar mass, M_{ICL} is the enclosed intracluster-light contribution when available, and T_{gate} is the support temperature supplied to the active-gate calculation.

The total baryonic seed mass is

$$M_b(< R_\Delta) = M_{\text{gas}}(< R_\Delta) + M_\star(< R_\Delta) + M_{\text{ICL}}(< R_\Delta). \quad (3)$$

When stellar or intracluster-light information is not available, the gas mass can be bracketed with a controlled stellar-loading factor,

$$M_b(< R_\Delta) = (1 + s_\star)M_{\text{gas}}(< R_\Delta), \quad (4)$$

with representative bracket values

$$s_\star = 0, \ 0.10, \ 0.20, \ 0.30. \quad (5)$$

The fixed baryonic closure anchor used by the cluster branch is

$$f_\pi = \frac{\pi}{20} = 0.157079632679. \quad (6)$$

The active response constants used in the current cluster implementation are

$$\Sigma_0 = 0.9960249674, \quad (7)$$

and

$$N_{C,0} = 0.1965088232. \quad (8)$$

The constant Σ_0 sets the near-closure response normalization. The constant $N_{C,0}$ sets the structural response-load normalization used by the active gate and by the structural mode.

2.1 Unit conventions and calculation constants

A direct calculation should use one coherent unit system. The formulas below are most convenient in SI units, followed by conversion of the final mass to solar masses. In the results tables of this paper, cluster masses are reported in units of $10^{14} M_\odot$, radii are reported in kpc, and temperatures are reported in keV.

Quantity	Value	Use in calculation
G	$6.67430 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$	Reference mass and acceleration
$M_\odot$	$1.98847 \times 10^{30} \text{ kg}$	Mass conversion
1 kpc	$3.085677581491367 \times 10^{19} \text{ m}$	Radius conversion
1 Mpc	$3.085677581491367 \times 10^{22} \text{ m}$	Hubble-unit conversion
m_p	$1.67262192369 \times 10^{-27} \text{ kg}$	Thermal support scale
1 keV	$1.602176634 \times 10^{-16} \text{ J}$	Temperature-energy conversion
μ	0.61	Intracluster plasma mean molecular weight
$H_0^{(\pi)}$	$69.4162507897 \text{ km s}^{-1} \text{ Mpc}^{-1}$	CPTG branch normalization
$H_0^{(\pi)}$ in SI	$2.24962747910 \times 10^{-18} \text{ s}^{-1}$	Use in SI formulas

Table 1: Constants and unit conversions used for standalone cluster calculations.

The Hubble normalization must be converted before substitution into the SI mass and temperature formulas:

$$H_0^{(\pi)}(\text{s}^{-1}) = H_0^{(\pi)}(\text{km s}^{-1} \text{ Mpc}^{-1}) \frac{1000 \text{ m km}^{-1}}{1 \text{ Mpc}}. \quad (9)$$

The virial support temperature is used in energy-temperature form. Thus a calculation using keV should compute $k_B T_{\text{vir},\Delta}$ as an energy and then divide by 1 keV:

$$T_{\text{vir},\Delta}(\text{keV}) = \frac{\mu m_p}{1 \text{ keV}} \left[\frac{\Delta}{2} H_C(z)^2 R_\Delta^2 \right]. \quad (10)$$

This is the same support scale represented by $k_B T_{\text{vir},\Delta} = \mu m_p (\Delta/2) H_C(z)^2 R_\Delta^2$. Using keV for both T_{gate} and $T_{\text{vir},\Delta}$ keeps the support ratio dimensionless.

2.2 Aperture and temperature rules

A valid hand calculation is aperture-consistent. The radius, baryonic masses, support temperature, and comparison mass must refer to the same declared aperture:

$$R_\Delta, \quad M_{\text{gas}}(< R_\Delta), \quad M_\star(< R_\Delta), \quad M_{\text{ICL}}(< R_\Delta), \quad T_{\text{gate}}, \quad M_{\text{obs},\Delta}. \quad (11)$$

This line represents one matched aperture state rather than a mixed set of measurements from different radii. For an R_{500} calculation, use R_{500} masses and the corresponding R_{500} comparison mass. For a ladder calculation, repeat the entire calculation separately at each chosen overdensity aperture.

The temperature gate is the thermal support temperature assigned to the selected aperture. In the X-COP R_{500} calculation reported here, $T_{\text{gate}} = T_{500}$. In a radial-profile or aperture-ladder diagnostic, T_{gate} should be the profile temperature or aperture support temperature associated with the same radial bin or aperture used to define the baryonic loading.

When an aperture radius is not supplied directly, it may be constructed from a comparison mass profile by solving the crossing condition

$$M_{\text{obs}}(< R_{\Delta}) = M_{\Delta, \text{ref}}(R_{\Delta}). \quad (12)$$

The baryonic masses are then evaluated or interpolated at that same radius before the active-gate calculation is performed.

The present calculation is a single-aperture cluster-response test. Strong mergers are not excluded from CPTG in principle, but they are excluded from the primary relaxed-aperture interpretation unless the gas, stellar, temperature, and mass components can be assigned consistently to the same dynamical aperture. In disturbed mergers, shocks, bulk flows, displaced gas, and non-equilibrium hydrostatic masses can turn the calculation into a merger stress test rather than a relaxed cluster-mass estimate.

3 Cluster Active-Gate Law

3.1 CPTG branch scale

The cluster reference scale is evaluated on the CPTG geometric branch. The branch expansion factor is

$$H_C(z) = H_0^{(\pi)} E_C(z), \quad (13)$$

with

$$E_C(z)^2 = \mathcal{B}_{\pi} + \mathcal{T}_{\pi}(1+z)^{\pi}. \quad (14)$$

The branch weights are

$$\mathcal{B}_{\pi} = 1 - \frac{1}{\pi}, \quad (15)$$

and

$$\mathcal{T}_{\pi} = \frac{1}{\pi}. \quad (16)$$

This scale supplies the geometric expansion and aperture reference used by the cluster calculation.

3.2 Reference mass and acceleration

At the selected aperture R_{Δ} , the CPTG reference mass is

$$M_{\Delta, \text{ref}} = \frac{\Delta}{2} \frac{H_C(z)^2}{G} R_{\Delta}^3. \quad (17)$$

This reference mass is the branch-native aperture mass scale. It supplies the denominator for the baryonic loading and the mass scale multiplied by the active-gate solution.

The corresponding reference acceleration is

$$g_{\Delta,\text{ref}} = \frac{GM_{\Delta,\text{ref}}}{R_{\Delta}^2} = \frac{\Delta}{2} H_C(z)^2 R_{\Delta}. \quad (18)$$

This acceleration is the local aperture acceleration implied by the same reference branch.

3.3 Baryonic loading

The baryonic loading variable is

$$F_{\Delta} = \frac{M_b(< R_{\Delta})}{M_{\Delta,\text{ref}}}. \quad (19)$$

The purpose of F_{Δ} is to measure how strongly the selected aperture is seeded by directly supplied baryonic matter relative to the CPTG branch-native reference mass. Larger values of F_{Δ} indicate a more heavily baryon-loaded aperture. Smaller values indicate a lighter baryonic seed relative to the branch reference scale.

3.4 Virial support temperature and support ratio

The aperture virial support temperature is most directly written as an energy scale,

$$k_B T_{\text{vir},\Delta} = \mu m_p \left[\frac{\Delta}{2} H_C(z)^2 R_{\Delta}^2 \right]. \quad (20)$$

For calculations using keV, this becomes

$$T_{\text{vir},\Delta}(\text{keV}) = \frac{\mu m_p}{1 \text{ keV}} \left[\frac{\Delta}{2} H_C(z)^2 R_{\Delta}^2 \right]. \quad (21)$$

Here μ is the mean molecular weight and m_p is the proton mass. This temperature-energy scale converts the branch-native aperture scale into a thermal support scale.

The dimensionless support ratio is

$$\theta_T = \frac{T_{\text{gate}}}{T_{\text{vir},\Delta}}. \quad (22)$$

The purpose of θ_T is to compare the supplied support temperature to the branch-native virial support temperature at the same aperture.

3.5 Critical support threshold and active gate

The critical support threshold is

$$\theta_{\text{crit}}(F_{\Delta}) = \left(\frac{F_{\Delta}^3}{\pi f_{\pi}^2 N_{C,0}} \right)^{1/2}. \quad (23)$$

This threshold converts baryonic loading into a gate threshold. A more heavily loaded aperture has a different support requirement than a lighter aperture.

The active gate is

$$Q_C = \frac{\theta_T}{\theta_{\text{crit}}(F_{\Delta})}. \quad (24)$$

The gate Q_C is the main response classifier. It compares the available support state to the loading-dependent critical support threshold. When Q_C is high, the aperture approaches closure. When Q_C is low, the aperture moves into a suppressed response state.

3.6 Active-gate mass ratio

The active-gate mass ratio y_{Δ} is solved from

$$y_{\Delta} = F_{\Delta} + (1 - F_{\Delta}) \frac{\left[1 + \left(\frac{y_{\Delta} N_{C,0}}{Q_C^2} \right)^{123/50} \right]^{-11/50}}{\Sigma_0}. \quad (25)$$

The first term, F_{Δ} , carries the direct baryonic loading. The second term carries the active curvature-response contribution. The kernel inside the brackets contains the structural load $y_{\Delta} N_{C,0}/Q_C^2$. This makes the mass ratio and structural mode self-consistent: the response depends on the load, and the load depends on the response.

The predicted CPTG cluster mass is

$$M_{\text{CPTG},\Delta} = y_{\Delta} M_{\Delta,\text{ref}}. \quad (26)$$

The corresponding aperture acceleration is

$$g_{\text{CPTG},\Delta} = y_{\Delta} g_{\Delta,\text{ref}}. \quad (27)$$

The practical use of y_{Δ} is direct. Once the active-gate equation is solved, y_{Δ} multiplies the reference mass to produce the predicted CPTG mass at the selected aperture.

4 Structural Mode $\mathcal{N}_{C,\Delta}$

The cluster structural mode is the dimensionless response-load variable already present in the active-gate kernel:

$$\mathcal{N}_{C,\Delta} = \frac{y_{\Delta} N_{C,0}}{Q_C^2}. \quad (28)$$

This quantity identifies where the aperture lies on the active curvature-response branch. Since Q_C appears in the denominator, closure-stable high- Q_C apertures carry lower structural load. Suppressed low- Q_C apertures carry higher structural load.

For comparison bookkeeping, it is useful to define the reciprocal closure measure

$$\mathcal{C}_{C,\Delta} = \frac{Q_C^2}{y_{\Delta} N_{C,0}} = \mathcal{N}_{C,\Delta}^{-1}. \quad (29)$$

The structural mode and reciprocal closure measure serve different descriptive purposes. $\mathcal{N}_{C,\Delta}$ increases as the aperture moves deeper into the active response load. $\mathcal{C}_{C,\Delta}$ increases as the aperture becomes more closure-stable.

The total-to-baryon mass multiplier can also be recorded as a separate diagnostic,

$$\mathcal{R}_{b,\Delta} = \frac{M_{\text{CPTG},\Delta}}{M_b(< R_{\Delta})}. \quad (30)$$

The structural mode used in this paper is $\mathcal{N}_{C,\Delta}$. The multiplier $\mathcal{R}_{b,\Delta}$ is a mass accounting ratio and is listed separately when needed.

Two X-COP R_{500} examples show the meaning of the structural mode. In the suppressed case A2319, the active gate is $Q_C = 0.6809$ and $y_{500} \simeq 0.9840$, giving

$$\mathcal{N}_{C,500} \simeq 0.4171. \quad (31)$$

In the closure-stable case ZW1215, the active gate is $Q_C = 1.6229$ and $y_{500} \simeq 1.0032$, giving

$$\mathcal{N}_{C,500} \simeq 0.0748. \quad (32)$$

Thus the suppressed aperture carries the larger structural response load, while the closure-stable aperture carries the smaller structural response load.

5 Numerical Solution and Gate Classes

The active-gate equation is solved by fixed-point iteration. Starting with

$$y_{\Delta,0} = 1, \quad (33)$$

update

$$y_{\Delta,n+1} = F_{\Delta} + (1 - F_{\Delta}) \frac{\left[1 + \left(\frac{y_{\Delta,n} N_{C,0}}{Q_C^2} \right)^{123/50} \right]^{-11/50}}{\Sigma_0}. \quad (34)$$

The iteration returns y_{Δ} , after which $M_{\text{CPTG},\Delta}$ and $\mathcal{N}_{C,\Delta}$ are computed.

The response classes used throughout this paper are

$$Q_C > 1.2 \quad \Rightarrow \quad \text{closure stable}, \quad (35)$$

$$0.8 \leq Q_C \leq 1.2 \quad \Rightarrow \quad \text{watch}, \quad (36)$$

$$0.5 \leq Q_C < 0.8 \quad \Rightarrow \quad \text{suppressed}, \quad (37)$$

and

$$Q_C < 0.5 \quad \Rightarrow \quad \text{strongly suppressed}. \quad (38)$$

These labels are not fitted classes. They are descriptive gate states produced by the active-gate variable.

6 Standalone Calculation Recipe

A cluster can be calculated directly from the preceding equations. The following recipe lists the required steps in the order a reader should apply them.

1. Select the aperture R_{Δ} and overdensity label Δ .
2. Convert R_{Δ} to meters and convert $H_0^{(\pi)}$ to s^{-1} .
3. Compute $E_C(z)$ and $H_C(z)$.
4. Compute $M_{\Delta,\text{ref}}$ and, if needed, $g_{\Delta,\text{ref}}$.
5. Compute the enclosed baryonic mass $M_b(< R_{\Delta})$.
6. Compute the baryonic loading $F_{\Delta} = M_b/M_{\Delta,\text{ref}}$.
7. Compute $T_{\text{vir},\Delta}$ in keV.
8. Compute $\theta_T = T_{\text{gate}}/T_{\text{vir},\Delta}$.
9. Compute $\theta_{\text{crit}}(F_{\Delta})$.
10. Compute $Q_C = \theta_T/\theta_{\text{crit}}$ and assign the gate class.

11. Solve the fixed-point equation for y_Δ .
12. Compute $M_{\text{CPTG},\Delta} = y_\Delta M_{\Delta,\text{ref}}$.
13. Compute the structural mode $\mathcal{N}_{C,\Delta} = y_\Delta N_{C,0}/Q_C^2$.
14. Compare $M_{\text{CPTG},\Delta}$ to an observed or inferred mass measured at the same aperture.

For hand calculation, the fixed-point iteration may be stopped once successive values of y_Δ no longer change at the desired precision. The numerical implementation used for the results in this paper used

$$y_{\Delta,0} = 1, \quad n_{\text{max}} = 500, \quad \epsilon = 10^{-13}. \quad (39)$$

6.1 Worked suppressed-aperture example: A2319 at R_{500}

A2319 provides a representative suppressed R_{500} calculation. The input row is

$$z = 0.0557, \quad \Delta = 500, \quad R_{500} = 1368 \text{ kpc}, \quad (40)$$

$$M_{\text{gas}} = 1.4556 \times 10^{14} M_\odot, \quad M_\star = 0.0695 \times 10^{14} M_\odot, \quad T_{\text{gate}} = 7.5495 \text{ keV}. \quad (41)$$

The intermediate calculation gives

$$M_b = 1.5251 \times 10^{14} M_\odot, \quad M_{500,\text{ref}} = 7.5942 \times 10^{14} M_\odot, \quad F_{500} = 0.2008. \quad (42)$$

The support and gate quantities are

$$T_{\text{vir},500} = 15.2050 \text{ keV}, \quad \theta_T = 0.4965, \quad \theta_{\text{crit}} = 0.7292, \quad Q_C = 0.6809. \quad (43)$$

Solving the active-gate equation gives

$$y_{500} = 0.9840, \quad M_{\text{CPTG},500} = 7.4726 \times 10^{14} M_\odot, \quad \mathcal{N}_{C,500} = 0.4171. \quad (44)$$

Against the same-aperture hydrostatic comparison mass $M_{\text{HSE},500} = 7.6781 \times 10^{14} M_\odot$, the ratio is

$$\frac{M_{\text{CPTG},500}}{M_{\text{HSE},500}} = 0.9732. \quad (45)$$

6.2 Worked closure-stable example: ZW1215 at R_{500}

ZW1215 provides a representative closure-stable contrast. The input row is

$$z = 0.0750, \quad \Delta = 500, \quad R_{500} = 1358 \text{ kpc}, \quad (46)$$

$$M_{\text{gas}} = 0.8106 \times 10^{14} M_{\odot}, \quad M_{\star} = 0.0426 \times 10^{14} M_{\odot}, \quad T_{\text{gate}} = 7.5906 \text{ keV}. \quad (47)$$

The intermediate calculation gives

$$M_b = 0.8532 \times 10^{14} M_{\odot}, \quad M_{500, \text{ref}} = 7.5839 \times 10^{14} M_{\odot}, \quad F_{500} = 0.1125. \quad (48)$$

The support and gate quantities are

$$T_{\text{vir}, 500} = 15.2962 \text{ keV}, \quad \theta_T = 0.4962, \quad \theta_{\text{crit}} = 0.3058, \quad Q_C = 1.6229. \quad (49)$$

Solving the active-gate equation gives

$$y_{500} = 1.0032, \quad M_{\text{CPTG}, 500} = 7.6082 \times 10^{14} M_{\odot}, \quad \mathcal{N}_{C, 500} = 0.0748. \quad (50)$$

Against the same-aperture hydrostatic comparison mass $M_{\text{HSE}, 500} = 7.6634 \times 10^{14} M_{\odot}$, the ratio is

$$\frac{M_{\text{CPTG}, 500}}{M_{\text{HSE}, 500}} = 0.9928. \quad (51)$$

7 ACCEPT Results

The ACCEPT application is a radial-profile diagnostic. ACCEPT provides Chandra-based intracluster-medium entropy and temperature information for a large archival cluster sample [1, 2]. The active-gate calculation was applied to clean outer-profile rows in order to test whether the formula orders profile apertures by response state.

For the ACCEPT profile pass, R_{out} is the outer radius of the selected profile row. The quantity $\Delta_{g, \pi}$ is the effective CPTG branch overdensity assigned to that row from the same reference-mass relation used throughout this paper,

$$\Delta_{g, \pi} = \frac{2GM_{\text{row}}}{H_C(z)^2 R_{\text{out}}^3}, \quad (52)$$

where M_{row} is the enclosed row-level mass proxy assigned to the selected ACCEPT profile row before the active-gate solution. It is the mass scale used to place the row

on the branch-overdensity coordinate $\Delta_{g,\pi}$. After that coordinate assignment, the active-gate calculation proceeds through the same variables used everywhere else in the paper: F_Δ , Q_C , and y_Δ . The profile-control quantity P_C is a dimensionless retained-row control factor for support-profile leverage. A larger P_C indicates that the selected outer row carries enough profile information to act as an active-gate diagnostic row. In the ACCEPT pass, P_C is used for row selection, while Q_C remains the physical gate variable that classifies the row.

The full clean-profile pass used the cuts

$$300 < \Delta_{g,\pi} < 800, \quad (53)$$

$$R_{\text{out}} > 0.5 \text{ Mpc}, \quad (54)$$

and

$$P_C > 0.25. \quad (55)$$

The pass produced 379 clean profile rows across 45 clusters. The resulting gate ordering is shown in Table 2.

Gate class	Rows	Clusters	Median F_Δ	Median Q_C	Median y_Δ
Closure-stable	173	18	0.2102	1.7704	1.0029
Watch	116	17	0.2434	0.9799	0.9997
Suppressed	85	16	0.2695	0.6691	0.9840
Strongly suppressed	5	2	0.2891	0.4871	0.9387

Table 2: ACCEPT full clean outer-profile active-gate behavior.

The ACCEPT full pass shows ordered behavior. Closure-stable rows have the largest median Q_C and remain near $y_\Delta = 1$. Suppressed rows have lower median Q_C and lower median y_Δ . Strongly suppressed rows sit at the lowest median Q_C and show the strongest reduction in y_Δ .

A representative 100-row diagnostic subset gives the same ordering pattern and is summarized in Table 3. This compact subset is useful as a smaller reproducibility and inspection sample.

Gate class	Rows	Clusters	Median F_Δ	Median Q_C	Median y_Δ
Closure-stable	44	15	0.2095	1.7444	1.0029
Watch	29	12	0.2359	0.9598	0.9994
Suppressed	22	11	0.2681	0.6426	0.9802
Strongly suppressed	5	2	0.2891	0.4871	0.9387

Table 3: ACCEPT representative 100-row active-gate diagnostic subset.

The ACCEPT result can be summarized as

$$Q_C \uparrow \Rightarrow y_\Delta \rightarrow 1, \quad (56)$$

and

$$Q_C \downarrow \Rightarrow y_\Delta < 1. \quad (57)$$

This is the expected gate-ordered response of the cluster active-gate law.

8 X-COP Results

X-COP provides X-ray and SZ-based cluster products designed to study cluster outskirts and hydrostatic mass profiles [3]. The CPTG active-gate calculation was applied to the available tested X-COP cluster products in two forms: a same-aperture R_{500} comparison and a six-aperture overdensity ladder.

8.1 Same-aperture R_{500} mass comparison

For each tested cluster, the R_{500} calculation used the same aperture for the supplied gas/baryonic input, the active-gate calculation, the predicted CPTG mass, and the hydrostatic comparison mass. The tested 12-cluster R_{500} pass is shown in Table 4.

Cluster	Q_C	Class	$M_{\text{CPTG},500}$	$M_{\text{HSE},500}$	CPTG/HSE
A2319	0.6809	Suppressed	7.4726	7.6781	0.9732
A2255	0.7826	Suppressed	5.1764	5.2569	0.9847
A3158	0.8350	Watch	4.2000	4.2637	0.9851
A2142	0.8772	Watch	8.8710	8.9523	0.9909
RXC1825	0.9520	Watch	4.0385	4.0827	0.9892
A3266	0.9568	Watch	8.7015	8.7968	0.9892
A85	0.9834	Watch	5.5848	5.6482	0.9888
A1644	1.0107	Watch	3.4432	3.4838	0.9884
A2029	1.0381	Watch	8.5874	8.6511	0.9926
A1795	1.0545	Watch	4.5819	4.6257	0.9905
A644	1.1520	Watch	5.8953	5.9408	0.9923
ZW1215	1.6229	Closure-stable	7.6082	7.6634	0.9928

Table 4: X-COP same-aperture R_{500} comparison. Masses are in $10^{14} M_\odot$.

The sample summary is

$$\text{median} \left(\frac{M_{\text{CPTG},500}}{M_{\text{HSE},500}} \right) = 0.9892, \quad (58)$$

$$\text{median} \left| \frac{M_{\text{CPTG},500}}{M_{\text{HSE},500}} - 1 \right| \simeq 0.0108, \quad (59)$$

$$\text{mean} \left| \frac{M_{\text{CPTG},500}}{M_{\text{HSE},500}} - 1 \right| \simeq 0.0119, \quad (60)$$

and

$$\text{max} \left| \frac{M_{\text{CPTG},500}}{M_{\text{HSE},500}} - 1 \right| \simeq 0.0268. \quad (61)$$

The R_{500} comparison therefore returns near-closure same-aperture masses across the tested X-COP sample. A2319 is the clearest suppressed R_{500} case in this table, while ZW1215 is the clearest closure-stable case.

8.2 X-COP structural-mode examples

The same R_{500} pass also illustrates the structural mode. Table 5 lists one suppressed case and one closure-stable case.

Cluster	Q_C	y_{500}	$\mathcal{N}_{C,500}$	Class
A2319	0.6809	0.9840	0.4171	Suppressed
ZW1215	1.6229	1.0032	0.0748	Closure-stable

Table 5: Representative X-COP R_{500} structural-mode examples.

The difference between these two rows is the structural content of the active gate. A2319 has a lower gate value and a higher response load. ZW1215 has a higher gate value and a lower response load.

8.3 Overdensity-ladder result

The X-COP products were also evaluated over an overdensity ladder,

$$R_{2500}, R_{2000}, R_{1500}, R_{1000}, R_{750}, R_{500}. \quad (62)$$

For each ladder aperture, R_Δ is found from the same branch-reference crossing condition,

$$M_{\text{HSE}}(< R_\Delta) = M_{\Delta,\text{ref}}(R_\Delta), \quad (63)$$

except at R_{500} when the catalog header value is supplied directly. The gas and stellar masses are then evaluated at the same radius, and the active-gate calculation is repeated independently for that aperture. In the ladder results reported here, the

support-temperature convention is fixed to the X-COP header value $T_{\text{gate}} = T_{500}$ for each cluster across the ladder. If an aperture-resolved temperature profile is supplied, the same formulas can instead use the profile temperature evaluated at the selected R_{Δ} . This makes the ladder an aperture-by-aperture structural response calculation.

The median aperture trend is shown in Table 6.

Aperture	Median Q_C	Median y_{Δ}	Median F_{Δ}	Median CPTG/HSE
R_{2500}	1.5781	1.0031	0.1250	1.0020
R_{2000}	1.4667	1.0029	0.1324	1.0019
R_{1500}	1.3063	1.0025	0.1366	1.0015
R_{1000}	1.1365	1.0016	0.1424	1.0001
R_{750}	1.0814	1.0011	0.1467	0.9996
R_{500}	0.9701	0.9995	0.1586	0.9892

Table 6: X-COP overdensity-ladder active-gate trend.

The median gate follows the ordered aperture trend

$$Q_C(R_{2500}) > Q_C(R_{2000}) > Q_C(R_{1500}) > Q_C(R_{1000}) > Q_C(R_{750}) > Q_C(R_{500}). \quad (64)$$

This is the strongest structural result in the current cluster pass. The inner apertures sit closer to closure. The outer apertures expose the active-gate separation. In the tested X-COP ladder, the median response moves from $Q_C = 1.5781$ at R_{2500} to $Q_C = 0.9701$ at R_{500} .

9 Cluster-Scale Interpretation

The cluster active-gate law produces three linked outputs: a predicted mass, a gate class, and a structural mode. The predicted mass is set by $M_{\text{CPTG},\Delta} = y_{\Delta} M_{\Delta,\text{ref}}$. The gate class is set by Q_C . The structural mode is set by $\mathcal{N}_{C,\Delta} = y_{\Delta} N_{C,0}/Q_C^2$.

The ACCEPT diagnostic demonstrates response ordering across radial profile rows. High- Q_C rows are closure-stable and remain near $y_{\Delta} = 1$. Low- Q_C rows suppress. This supports the use of Q_C as the practical ordering variable for cluster profile apertures.

The X-COP same-aperture comparison demonstrates that the active-gate law can be evaluated against real cluster hydrostatic products at R_{500} . The tested sample returns near-closure masses with percent-level median differences relative to the supplied hydrostatic comparison masses.

The X-COP aperture ladder demonstrates that the gate is not merely a single-aperture mass normalizer. The median Q_C changes systematically with aperture.

Inner apertures are more closure-stable; outer apertures show stronger active-gate sensitivity. This is the desired behavior for an aperture-resolved cluster law.

The cluster-scale response can therefore be summarized as

$$\text{baryonic loading} + \text{thermal support} + \text{aperture scale} \quad (65)$$

$$\longrightarrow \text{active curvature response.} \quad (66)$$

The structural mode supplies the additional diagnostic needed to describe the state of that response branch:

$$\mathcal{N}_{C,\Delta} \uparrow \Rightarrow \text{deeper active response load,} \quad (67)$$

and

$$\mathcal{N}_{C,\Delta} \downarrow \Rightarrow \text{closure stable response state.} \quad (68)$$

10 Conclusion

This paper presents a CPTG active-gate law for galaxy clusters. The law evaluates a selected overdensity aperture using baryonic loading, thermal support, the CPTG branch scale, and a structural response-load variable. The output is a predicted aperture mass, an active-gate class, and a structural mode.

The formula is compact. The baryonic loading F_Δ supplies the aperture seed. The support ratio θ_T compares the supplied temperature gate to the branch-native virial support temperature. The gate Q_C determines the response state. The fixed-point solution gives the mass ratio y_Δ . The structural mode $\mathcal{N}_{C,\Delta}$ identifies the load carried by the active curvature-response branch.

The ACCEPT diagnostic shows clean gate ordering across outer radial-profile rows. The X-COP R_{500} comparison gives near-closure same-aperture masses across the tested 12-cluster sample, with median CPTG/HSE approximately 0.989 and median absolute fractional difference approximately 1.08%. The X-COP overdensity ladder shows a continuous structural transition from inner closure at R_{2500} to outer active-gate sensitivity at R_{500} .

The central cluster-scale result is that CPTG can be expressed as an aperture-resolved active-gate calculation. In this form, cluster mass prediction and cluster structural response are not separate procedures. They are two outputs of the same baryonic-loading, support-temperature, and curvature-response law.

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